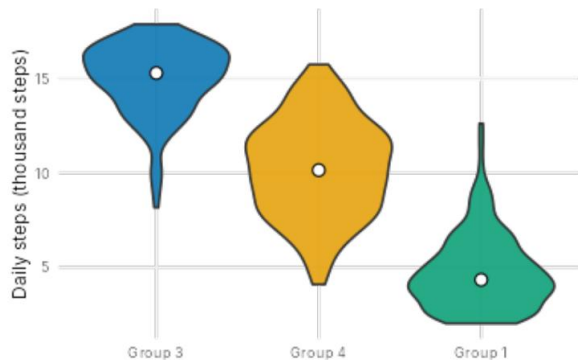


R.No: 30 Reg. No: 192524110 Name: Pranayaa S

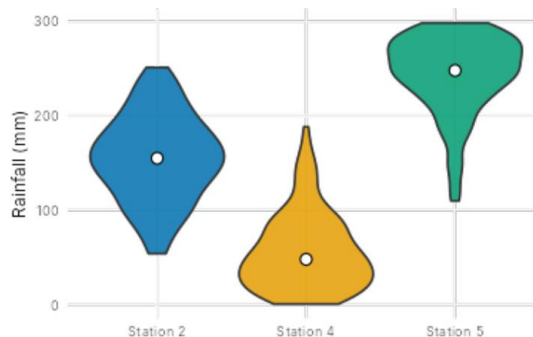
Part A (18 marks). Each chart is a violin plot (white dot = median) made in R with ggplot2. For every violin circle the skew (1 mark) and the relation of the mean to the median (0.5 mark).

Chart 1: Daily steps (thousand steps)



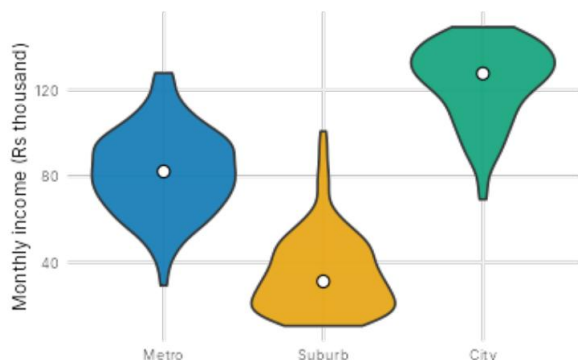
Violin	Skew (circle one)	Mean vs Median (circle)
Group 3	Right / <u>Left</u> / Symm.	Mean > / <u>≤</u> / ~ Median
Group 4	Right / Left / <u>Symm.</u>	Mean > / < / ~ <u>Median</u>
Group 1	<u>Right</u> / Left / Symm.	<u>Mean ></u> / < / ~ Median

Chart 2: Rainfall (mm)



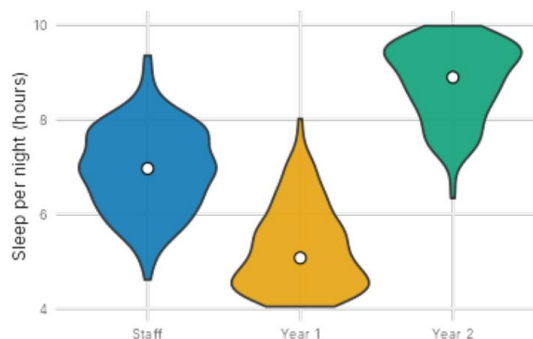
Violin	Skew (circle one)	Mean vs Median (circle)
Station 2	Right / Left / <u>Symm.</u>	Mean > / < / ~ <u>Median</u>
Station 4	<u>Right</u> / Left / Symm.	<u>Mean ></u> / < / ~ Median
Station 5	Right / <u>Left</u> / Symm.	Mean > / <u>≤</u> / ~ Median

Chart 3: Monthly income (Rs thousand)



Violin	Skew (circle one)	Mean vs Median (circle)
Metro	Right / Left / <u>Symm.</u>	Mean > / < / ~ <u>Median</u>
Suburb	<u>Right</u> / Left / Symm.	<u>Mean ></u> / < / ~ Median
City	Right / <u>Left</u> / Symm.	Mean > / <u>≤</u> / ~ Median

Chart 4: Sleep per night (hours)



Violin	Skew (circle one)	Mean vs Median (circle)
Staff	Right / Left / <u>Symm.</u>	Mean > / < / ~ <u>Median</u>
Year 1	<u>Right</u> / Left / Symm.	<u>Mean ></u> / < / ~ Median
Year 2	Right / <u>Left</u> / Symm.	Mean > / <u>≤</u> / ~ Median

Part B (2 marks)

B1. In Chart 2, violin "Station 4" is not symmetric. Towards which side is its tail, and which is the better measure of a typical rainfall here: mean or median? Give one reason. [1]

Answer: The tail extends toward the upper (high-rainfall) side, so the violin is right-skewed. The median is the better measure, because a few very wet days pull the mean upward while the median is not affected by them.

B2. Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown. [1]

Answer:

```
ggplot(df, aes(x = group, y = value)) + geom_violin() + stat_summary(fun = median, geom = "point", colour = "white", size = 3)
```